

Course Syllabus: *Segnali Elettrofisiologici*

Master's Degree in Bioengineering | UNIMORE

- **Academic Year:** 2026/27
- **Course code:** BIOING_1 - 15806
- **Course info:** year 1, winter semester, 6 credits
- **Lecturer:** Prof. Eng. Michele GIUGLIANO, PhD
- **Format:** In-person lectures, chalk-and-talk, computational notebooks
- **Attendance:** Very strongly recommended

Syllabus: It is an outline of the topics taught in the course. It also contains additional information, designed to 1) preemptively answer students' questions, 2) list policy and mutual expectations and 3) the consequences of failing to meet them. It does not replace the official information about the course, published online at <https://unimore.coursecatalogue.cineca.it>, but extends it.

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1. Learning Objectives

Students will be exposed to a selection of elementary topics in the genesis of electrical signals in nervous tissue, the biochemical and molecular signals of excitable cells. Students will understand the rationale behind a mechanistic dissection of the functioning of the mammalian nervous system at the cellular, microcircuit, and population levels. Students will ultimately gain expertise in this area and be able to apply their knowledge to capture an in-depth quantitative component in cutting-edge scientific publications.

Students will begin to make *quantitative* judgments and evaluations in the fields of *Electrophysiology* and Cellular Biophysics, analysing how significant progress can be made in these areas and where the availability of solid theoretical tools has proven extremely fruitful. Students will become fluent in this progression and skill.

Students will be introduced to a quantitative style of neural systems biophysics as an opportunity to expand their learning capabilities beyond “*enumeration*” in biology and into a new “*synthesis*” and “*analysis*” approach.

2. Prerequisite knowledge

Prerequisite for this course: By enrolling in this Master's Degree program, students have formally declared that they satisfy the UNIMORE curricular requirements. The requirements that students declare they already know constitute exactly the foundational knowledge requested for this course. This is spelled out explicitly below and listed as A, B, and C, with specific regards to some mathematical and scientific aspects:

A. Mathematics & Statistics

- **Basic Math, Functions & Algebra:** Elementary math, Algebra, Linear and nonlinear algebraic equations, functions of one variable, graph of a function, linear functions, exponentials, natural logarithms, and sinusoids/cisoids.
 - **SSD Reference:** MAT/05 (Analisi Matematica)
- **Calculus & Vector Analysis:** Derivatives (including composite functions/chain rule), definite and indefinite integrals, limits, first-order Taylor series expansion, partial derivatives, and spatial gradients; complex numbers, vector calculus.
 - **SSD Reference:** MAT/05 (Analisi Matematica)
- **Differential Equations:** First-order ordinary differential equations (ODEs) with constant coefficients, solving via homogeneous and particular solutions (forcing terms), and boundary/initial conditions (the Cauchy problem). Conceptual understanding of partial differential equations (PDEs).
 - **SSD Reference:** MAT/05 (Analisi Matematica) / MAT/07 (Fisica Matematica)
- **Linear Systems Theory:** The Dirac delta function, convolution integrals, impulse responses, and the fundamental concepts of low-pass and high-pass filters, Fourier transform, Laplace transform.
 - **SSD Reference:** ING-INF/03 (Telecomunicazioni) / MAT/05 (Analisi Matematica)
- **Probability & Stochastic Processes:** Probability, random variables, density distributions, stochastic processes, random fluctuations/noise.
 - **SSD Reference:** SECS-S/01 (Statistica) / MAT/06 (Probabilità)
- **Numerical Methods:** Discretization of continuous space and time, numerical integration methods (e.g., Euler, Runge-Kutta), and the concept of numerical stability.
 - **SSD Reference:** MAT/08 (Analisi Numerica)

B. Physics & Biophysics

- **Classical Mechanics & Thermodynamics:** Newton's law, mass, work, energy, conservative force fields, friction (Stokes' law), and the concepts of

thermodynamic equilibrium and diffusion.

– **SSD Reference:** FIS/01 (Fisica Sperimentale)

- **Electrostatics & Electromagnetism:** Coulomb's force, vector vs. scalar fields, electrostatic potential, electric fields, charge quantization, permittivity of a vacuum, relative permittivity of a medium, and the superposition principle.

– **SSD Reference:** FIS/01 (Fisica Sperimentale)

- **Circuit Theory & Electronics:** Ohm's law, electrical resistance/conductance, capacitance, properties of parallel-plate capacitors, RC circuits, time constants (τ), Kirchhoff's Current and Voltage Laws, and lumped vs. distributed parameter models.

– **SSD Reference:** FIS/01 (Fisica Sperimentale) / ING-INF/01 (Elettronica)

C. Chemistry & Physical Chemistry

- **Solutions & Stoichiometry:** Solutes vs. solvents, moles, molarity, Avogadro's number, conversion between volumetric concentration, surface density, and absolute number of particles.

– **SSD Reference:** CHIM/03 (Chimica Generale e Inorganica)

- **Ionic Species & Electrochemistry:** Properties of Na^+ , K^+ , Cl^- , and Ca^{2+} , electrochemical gradients, pH, and the dissociation of salts in aqueous media.

– **SSD Reference:** CHIM/03 (Chimica Generale e Inorganica)

- **Chemical Kinetics:** Reaction rates, reversible vs. irreversible reactions, the Law of Mass Action, and chemical equilibrium.

– **SSD Reference:** CHIM/03 (Chimica Generale e Inorganica) / FIS/03 (Fisica della Materia)

Explicit review of prerequisite material in class: Although the very prerequisites for the enrollment in this Master's Degree program already cover the prerequisite for **this** course, the Lecturer is well aware of the diverse academic backgrounds of the students cohort and of their heterogeneous strengths and weaknesses. For such reason **the Lecturer will anyway explicitly review all the prerequisite material, during classes**. Please note that basic math, arithmetics, algebra, and calculus are covered through an asynchronous blended learning module ("Chapter 0") available online (<https://youtube.com/playlist?list=PL4ziDQ1W-M22rbHT8dfz4IDmAp3z4dBuq>) for all students and through a series of online (asynchronous) learning resources, presented in the overheads on Day 1 class.

Students with lack of prerequisite knowledge: Given the interdisciplinary nature of Bioengineering, students may occasionally discover significant gaps in their foundation knowledge throughout the classes. If a student perceives that the course requires prerequisites that feel too difficult, they are strongly encouraged to contact the Lecturer as early as possible. While these concepts are part of the knowledge formally declared by the student as a prerequisite for accessing

the Master in Bioengineering for Innovation in Medicine—and thus the Lecturer is not, in principle, responsible for reteaching them—the Lecturer is nevertheless fully committed to gladly supporting the student’s learning. However, **this assistance can only be provided if these challenges are communicated proactively**. If the Lecturer is not made aware of specific deficiencies in advance, it will be impossible to offer the necessary support or tailored guidance later in the course.

Day 1, diagnostic non-graded questionnaires: To help the Lecturer calibrate the subsequent review of prerequisites during the forthcoming lectures, an anonymous Diagnostic Questionnaire will be provided electronically, during the first class. It is ungraded and it samples all the above topics. If a student chooses not to complete it in class (or if they join the course late and fail to report to the Lecturer their own weaknesses in the prerequisite material), the Lecturer will proceed with the pacing of subsequent lectures under the assumption that a satisfactory mastery of all foundation mathematics, physics, and chemistry is already present.

3. Core Class Content

0. Mathematical Preliminaries Mathematical functions and their graphs Derivatives and Integrals The Dirac’s Delta and the Convolution Integral Ordinary differential equations and their solution Numerical solutions and the Euler forward method

1. Introduction Definition of “biological signal” Scientific motivations and innovation in medicine and neurobiology Examples of electrophysiological signals of the nervous system

2. EEG, Extracellular, and Intracellular Electrophysiological Signals Mentions of EEG and its applications in neurology Origins of EEG and its counter intuitive aspects

3. Biophysical Preliminaries Definition of ionic mobility in solution Capacitive and resistive properties of biological membranes Electrical Circuit Equivalent Models Definition of ionic fluxes and introduction of drift and diffusion

4. Neuroelectronics Thermodynamic equilibrium and the Nernst potential Transport currents through a semipermeable membrane Ohmic approximation to membrane permeability Goldman’s equation and non-ohmic approximation Goldman-Hodgkin-Katz equations Definition and inference of “resting” membrane potential

5. Introduction to Cell Excitability Charge-balance equation Thevenin electrical equivalent circuit of a permeable membrane Intuitive account for permeability changes through time Introduction to Action Potentials (APs)

- 6. Further biophysical preliminaries** Mass-action law, chemical reactions, and equations Kinetic schemes Non-deterministic microscopic interpretations of kinetic schemes
- 7. Excitability according to Hodgkin & Huxley** In-depth description of the HH model for AP generation Markov kinetic schemes for voltage-gated ionic channels Non-deterministic correlates of cell excitability
- 8. Synaptic transmission: chemical and electrical synapses** Kinetic schemes for ligand-gated synaptic receptors Classification of synaptic receptors Excitatory and inhibitory synapses Electrical synaptic transmission and connectivity Reduced/simplified model of chemical synaptic transmission
- 9. Synaptic plasticity** Short-term, homosynaptic plasticity Short-term depression and short-term facilitation Kinetic schemes for dynamics of ready-releasable synaptic resources Tsodyks-Markram model and its reduced version Long-term, heterosynaptic plasticity Spike-timing dependent plasticity, triplets of APs Considerations related to connectivity motifs
- 10. Extracellular Electrophysiological Signals** Intra- and extracellular potential and spatially-distributed properties Derivation of cable equation and Rall’s intuition Definition of electrotonus, solution of the cable equation Significance for spatial attenuation and temporal filtering of synaptic potentials Back-propagating action-potentials Synaptic Democracy
- 11. Volume Conductor Theory and Extracellular Potentials** Point current sources and line current sources Contributions to extracellular potentials Discussion on applications of Volume Conductor Theory to common electrophysiological signals: spikes and LFPs
- 12. Analysis of Electrophysiological Signals** Signals low-pass and band-pass filtering Peak-detection and spike-sorting Features extraction from intracellular signals Elementary spike-train analysis Brief mentions to Information Theory, Entropy, and Mutual Information for spike-train analysis

4. Study Materials, Textbooks, Resources

- **Primary References:** Selected chapters from Hanes et al., (2024) “Electric Brain Signals”; and selected chapters from Johnston & Wu, 1995 “Foundations of Cellular Neurophysiology”
- Students with significant gaps in their foundational knowledge, are recommended consulting these additional reference material: *Weiss TF (1996) “Cellular Biophysics” vol. 1, MIT Press.*; *Molecular Biology of the Cell* (Alberts et al.); *From Neuron to Brain* (Nicholls et al., 1996); *Principles of Neural Science* (Kandel et al.).
- **Readings:** Some (compulsory) reading material, such as *Eye, Brain, and Vision* (Hubel) and a condensed “Bignami-style” summary document on

basic neuron and synapse physiology, are available on GitHub.

- **Online Resources and Study Material:** The course has a a) GitHub repository and b) a dedicated website for distributing educational materials and supporting students:
 - 1) The **Course Website** (<https://mgiugliano.github.io/ePhysSignals>) provides a structured overview of the course, including the learning objectives, the updated class schedule, and a detailed list of all the topics covered. In the “Resources” section, links to video tutorials, scientific literature, additional readings, mathematical prerequisites are provided.
 - 2) The **GitHub Repository** (<https://github.com/mgiugliano/ePhysSignals>) serves as a central archive where lecture slides, Python notebooks, and reading materials are stored. Students must refer to this material to support their study and dive deeper into the technical and practical aspects of the module. The repository also features the refined (LLM-curated) transcripts of last year(s)’ classes. ****Please keep in mind that these are generated in an unsupervised manner using Large Language Models (LLMs) and with no guarantees of correctness or completeness of this content.**
 - 3) **Anki Flashcards:** Assuming students might be Anki’s users (i.e., a spaced repetition flashcard program), in the GitHub repository of the course there are over 1,700 flashcards covering the course chapters to support their study. ****Please keep in mind that these are generated in an unsupervised manner using Large Language Models (LLMs) and with no guarantees of correctness or completeness of this content.**
 - 4) A **Quiz Web App** (<https://mgiugliano.github.io/ePhysSignals/quiz>) is also offered to the students, featuring a very large database of typical exam/quiz questions (available only in Italian). Please keep in mind that these are generated in an unsupervised manner using Large Language Models (LLMs) and with no guarantees of correctness or completeness of this content.
 - 5) A **NotebookLM Study Guide** has been tentatively prepared and offered to the students to assist with interacting with the course material and querying concepts. **Please keep in mind that this is generated in an unsupervised manner using Large Language Models (LLMs) and with no guarantees of correctness or completeness of this content.**

“LLM Debugging” Challenge: As STEM scholars, students’ duty is the rigorous challenging of information received, not its passive acceptance. The LLM-curated transcripts, the LLM-generated Anki flashcards and quizzes are unsupervised and *may* occasionally contain *hallucinations*, contradictions or errors. The Lecturer calls all the students to treat these materials as raw, unreviewed literature. When you suspect a LLM hallucination, you are invited to formally “debug” it. To do so, you must submit a concise, written refutation

to the Lecturer via email, that cites the context, isolates the specific error, and corrects it using first-principles reasoning and explicit, page-level citations from the book references (e.g., *Halnes et al.* or *Johnston & Wu*). **The official textbooks and the Lecturer's in-class instructions remain the sole authoritative sources of truth for exam preparation.**

5. Grading Procedures

- **Overall/compound grading:** The grade for *this* module will be weighted by its credits (i.e., 6 CFUs) and combined with the grades the student achieves in all other modules. This determines the **final grade**, after rounding to the *closest integer*. Students must pass anyway each module (i.e., 18/30 minimum). 30, *Cum laude* is awarded only if, after the CFU-weighted average and rounding, the grade is 32/30.
- MG checks once a month (during the exams periods only, i.e. Jan-Feb, Jun-Jul, Sep) for the necessary conditions for formal registration of the **full** score, averaging across all the individual modules. These are: every other lecturer has officially registered your score and has officially communicated it to me (beware of the “grace period” to reject a score, as it introduces a delay in the entire procedure). Do NOT email MG to demand any urgent registration of your full score.
- Grades achieved by a student in any given module “**expires**” after **24 months**, requiring the students to pass that module again.
- **Grading for this module:** The assessment of student performance in *this module* is designed to evaluate both continuous engagement with the material and ultimate mastery of the entire curriculum, structured around three distinct components:
 - 1) a **written test** (2h, no books, no notes, questionnaire closed-answers, exercises) - (max **27 points**)
 - 2) a series of **brief, continuous in-class assessments** - (max **3 points**), and
 - 3) an **optional** oral exam, as a 20-minute blackboard interview, necessary to achieve **30 e lode (cum laude)**. **Warning:** This is a comprehensive evaluation, where poor performance may possibly *decrease* the “starting” grade. .

In-class assessments take the form of brief digital quizzes, exclusively administered in-class every week. Missed quizzes cannot generally be recovered or compensated for. However, standard university exceptions will be made for documented medical emergencies or *cause di forza maggiore*. Furthermore, students who are formally unable to attend classes (e.g., certified working students) must contact the Lecturer at the beginning of the semester to agree upon an alternative

equivalent assessment, ensuring they retain the opportunity to achieve the maximum score.

Active LLM Debugging (Bonus): To reward the intellectual rigor and time required by the “LLM Debugging” Challenge, students who successfully submit a formal, textbook-backed refutation of a factual biophysical error found within the AI-generated course materials will be awarded up to **+1 bonus point**, which is strictly additive to their final exam grade. This bonus is offered on a strict first-come, first-served basis: if multiple students spot the same error, only the first to report it receives the bonus point.

- **Important Dates:** Exams sessions date (in January, February, June, July, September) are published on esse3.unimore.it. *Note: Students must register at least 10 days in advance for any exam session. Given that this module grants 6 CFUs, students should consider prioritizing this exam over those with a lower number of CFUs.*
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6. Logistics and Policies

- **Class start and Breaks:** Lectures begin promptly **on the hour**. The *academic quarter* is not observed. Lectures breaks for **10 minutes every 50 minutes**.
- **Overheads:** Access to the updated overheads, used in class on a given day, is made available on GitHub (<https://github.com/mgiugliano/ePhysSignals>) not less than 30 minutes before each class. Students may then focus on oral and chalk presentation by the Lecturer and on the interactions. Overheads are a *support* for the knowledge transfer not an end by itself. Contact hours in class serves to extend, explain, and integrated the information that may be referred to in the slides.
- **Videotaped lectures:** As the Master’s Degree program in Bioengineering is **not** an “online degree”, live lectures are neither systematically recorded nor streamed, unless noted otherwise. No provision for hybrid or online attendance is offered.
- Supporting working students, unable to attend the classes, or those dealing with severe and prolonged illness, the archival video recordings of past year(s)’ lectures are made available on the Lecturer YouTube channel (https://www.youtube.com/@michele_giugliano). Those older recordings are provided as a courtesy, with no implied guarantee of any correspondence in content, structure, or schedule with the current academic year’s classes. Thus, relying on those videotaped lectures as a substitute for in-person attendance is non-ideal, it is discouraged, and done entirely at the student’s own responsibility.
- **Questions:** Students are encouraged to actively ask questions and vig-

orously participate intellectually **during the class**. As they are early in their training career, students are definitely allowed to ask *stupid* questions. Overly shy students may approach the Lecturer before the start of a class or during the breaks for quick questions, being aware they are preventing their peers to benefit from the “question and answer”. The Lecturer is passionate about passing his knowledge and fully committed to answer all questions at any time, including “Office Hours” (see Section 8).

- **Attendance Policy:** While class attendance itself is not compulsory, students are very strongly recommended to attend each class in person, to be on time and to avoid leaving before the end. However, if you are present in class, **attendance tracking is compulsory**. This is because attendance records are strictly tied to the assessment of the weekly in-class quizzes (see the Grading Procedures). Note: the Lecturer does **not** accept submission of the in-class questionnaires by those students who are not physically tracked in class.
- Attendance is tracked by the Lecturer for each class via the UNIMORE mobile app (<https://www.unimore.it/it/servizi/unimore-app>). Students **must** install this app on their mobile devices and use it to record their presence when they are in class. The Lecturer provides a unique code at the start of each class (e.g., 9PSTK) to be logged on the UNIMORE mobile app. Students exceptionally arriving late and thus *missing* the unique code communication must immediately alert the Lecturer.
- **Device Policy and Classroom Engagement:** While students will need their smartphones briefly to record attendance via the UNIMORE app and participate in the electronic in-class quizzes, laptops, tablets, and smartphones are otherwise strongly discouraged during class to minimize distractions and promote active engagement. The Lecturer does not normally request that students leave their bags and devices at the class entrance, unless for the written exam. Devices inherently bring hundreds of other distractions that are always just one tap or one notification away.
- Students are requested to seriously commit to active participation in each class, demonstrating an investment of attention and focus that matches the Lecturer’s own investment of energy, effort, and care in delivering the lecture.
- **On Taking Notes:** Taking handwritten notes is strongly encouraged, as it forces real-time synthesis of the material. Relying solely on the Lecturer’s overheads is insufficient for passing the exam.
- **Academic Honesty** Academic dishonesty, plagiarism, and cheating are serious violations of university trust and will result in immediate administrative consequences
- **Visit to the Lab:** Interested students are offered, upon appointment, a short tour of the Lecturer’s labs.

- The Lecturer can also be contacted at any times for information on stages, internships, and master thesis projects.
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7. Teaching Style, Language, and Philosophy

- **Format:** The teaching activity consists of frontal lectures where theoretical aspects and application examples are introduced and discussed. Furthermore, exercises and numerical computer simulations are presented in-class.
 - **Language:** The Lecturer **prefers to deliver this course entirely in English and he offers it as an option.** This offer becomes a compulsory choice in the presence of *even a single international student* , to ensure full, open access to global knowledge.
 - Lecturing in English transforms the classroom into a realistic professional microcosm, offering students a protected, low-risk environment to naturally absorb (and practice) technical terminology before entering the competitive high-tech industry or international research sectors. The Lecturer's proposal to use spoken English during classes is an offer of a strategic investment, designed to bridge the gap between *local* education and the *global* job market.
 - When no international students are enrolled in the course, students **vote** on the language of the course during the first class: Italian or English. Students' choice carries definitive implications.
 - The university's Erasmus+ bilateral agreements and international placements thrive on a strong foundation of English proficiency. Students are asked to consider that opting for the English track significantly facilitates the Lecturer's ability to personally sponsor and confidently recommend them for international internships within his global research network. If the Italian track is chosen, students may need to rely more heavily on the networks of other UNIMORE faculty members for international placements.
 - **Language used by students:** Regardless of the chosen lecture language, students can ask questions or seek clarifications speaking either English or Italian during class or office hours. They similarly retain full freedom to choose either Italian or English for their (optional) oral exam.
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8. Contact and Communications

- **Contact:** Students can reach the Lecturer through email (michele.giugliano@unimore.it) or instant messaging (Teams) via

<https://teams.microsoft.com/l/chat/0/0?users=michele.giugliano@unimore.it>

- **Office hours:** The Lecturer provides explanations on the topics covered in the course **only in person, and upon appointment**, and not via Teams or email.
- Students should not hesitate to contact the Lecturer for explanations on any aspect of the course, or to request additional study material and pointers to the literature.
- Students are invited to use Teams to post their questions publicly, so that the entire class may try to answer them, with all benefiting from each interaction.
- **Teams App:** The only official channel used by the Lecturer to communicate to the students is the dedicated **Microsoft Teams group**. Teams is where announcements, posts, information, links, papers, and last-minute or urgent alerts are communicated to the students for their own benefit, interest, as well as for the administrative, logistic, and pedagogical aspects of the course.
- Students enrolled in this course should have been automatically added to the Teams group members by UNIMORE IT services. Students without a UNIMORE email account must **urgently** request the Lecturer, in written (email), to be added to Teams (i.e., *manually*).
- Students should install the Teams app on their device(s) and enable notifications.
- Teams is **not** meant as a unidirectional or passive communication channel. It serves as a **forum** where, **as a community**, students interact with the Lecturer and among themselves by asking/answering each others questions and exchanging information.
- **Student Responsibilities:** Any last-minute communication, sudden schedule changes, or emergency announcements regarding classes and exams will be posted by the Lecturer **exclusively** on Teams. It is entirely students' responsibility to stay informed.
- Missing an urgent update due to unchecked notifications will **not** be accommodated.
- Throughout their **entire** academic tenure as students at UNIMORE, individuals **must** check their Teams app and their unimore.it email account for new posts and messages **every working day, at least once per day**.